CLASS 6 — MATHS · FORMULA SHEET

Chapter 5: **Prime Time** | Quick revision, one page

Factors and multiples

- A factor of n divides n exactly; the smallest factor is 1 and the largest is n.
- A multiple of n is $n \times 1, n \times 2, n \times 3, \ldots$ Every number is a factor and a multiple of itself.
- Prime: exactly two factors (1 and itself), e.g. 2, 3, 5, 7, 11. Composite: more than two factors. 1 is neither.

Divisibility tests

- By 2: last digit 0, 2, 4, 6, 8. By 5: last digit 0 or 5. By 10: last digit 0.
- By 3: digit sum divisible by 3. By 9: digit sum divisible by 9.
- By 4: last two digits divisible by 4. By 8: last three digits divisible by 8.
- By 11: (sum of digits at odd places) $-$ (sum at even places) is 0 or a multiple of 11.

Prime factorisation, HCF and LCM

- Every number > 1 is a product of primes, e.g. $60 = 2^2 \times 3 \times 5$.
- HCF: product of the common prime factors with the smallest powers.
- LCM: product of all prime factors with the greatest powers.
- For two numbers: $\mathrm{HCF} \times \mathrm{LCM} =$ product of the two numbers.
- Co-prime numbers: HCF $= 1$ (e.g. 15 and 28).

Sieve of Eratosthenes

- Cross out 1, then multiples of 2, 3, 5, 7, ... (keep the prime itself). Primes up to 30: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.

Problem types

- Meet again together $\Rightarrow$ LCM. Largest equal pieces or tiles $\Rightarrow$ HCF.

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